

Graph attributes and performance measures

Supply Chain Network Graph Attributes		
Sr	Attribute/Feature	Description
1	Number of nodes	Total number of nodes in the SC network
2	Number of edges	Total number of links in the SC network
3	Number of tiers	Number of echelons between raw material supply and the final customer
4	Type of SC	Type of supply chain (Forward, Reverse, Closed-loop)
5	Demand	Demand faced by the network, given rather than chosen
6	Demand Variability	Variability of demand across the network, such as its distribution or seasonality
Supply Chain Network Graph Performance Measures		
Sr	Performance Measure	Description
1	Supply chain cost	The overall cost of the supply chain in a given time period/time unit (month/quarter/half year/year)
2	Average SC cost	Average supply chain cost per time unit (daily/monthly/quarterly)
3	Supply chain net profit	Net revenue generated by the supply chain in a given time period, or a financial return such as net present value, internal rate of return, or return on investment
4	Average SC net profit	Average net revenue generated by the SC per time unit
5	Customer service level	Percentage of demand met on time in a given time period
6	Resilience level	Overall resilience of the network under disruption, based on node and edge reliability
7	Time to recover	Time the network takes to return to its pre-disruption performance after a disruption
8	Average cost per item	Average cost incurred per unit of product (ratio of total cost to received demand)
9	Average cost per order	Average cost of an order
10	Inventory carry cost	Total cost of storing each product item at nodes in the network
11	Inventory spend cost	Total expenses for replenishing the inventory at nodes
12	Transportation cost	Total transportation cost (sum) across nodes in the network
13	Revenue	Total income generated by all nodes in the network
14	Availability	Supply availability rate (percentage of demand nodes that have access to supplies)
15	Connectivity	Size of the largest functional subnetwork (where there is a path between any two nodes and at least one supplier exists)
16	Accessibility	Based on average supplier-to-consumer path length
17	Flow complexity	Total number of forward, reverse, and within-tier material flows
18	Density	Average number of nodes per geographic cluster
19	Density variance	Variance in node density across clusters
20	Supplier complexity	Percentage of nodes that act as suppliers in the supply chain network
21	Node criticality	Percentage of critical nodes (nodes that could cut the main flow when down)
22	Bullwhip ratio	Ratio of the variance of orders placed upstream to the variance of demand downstream